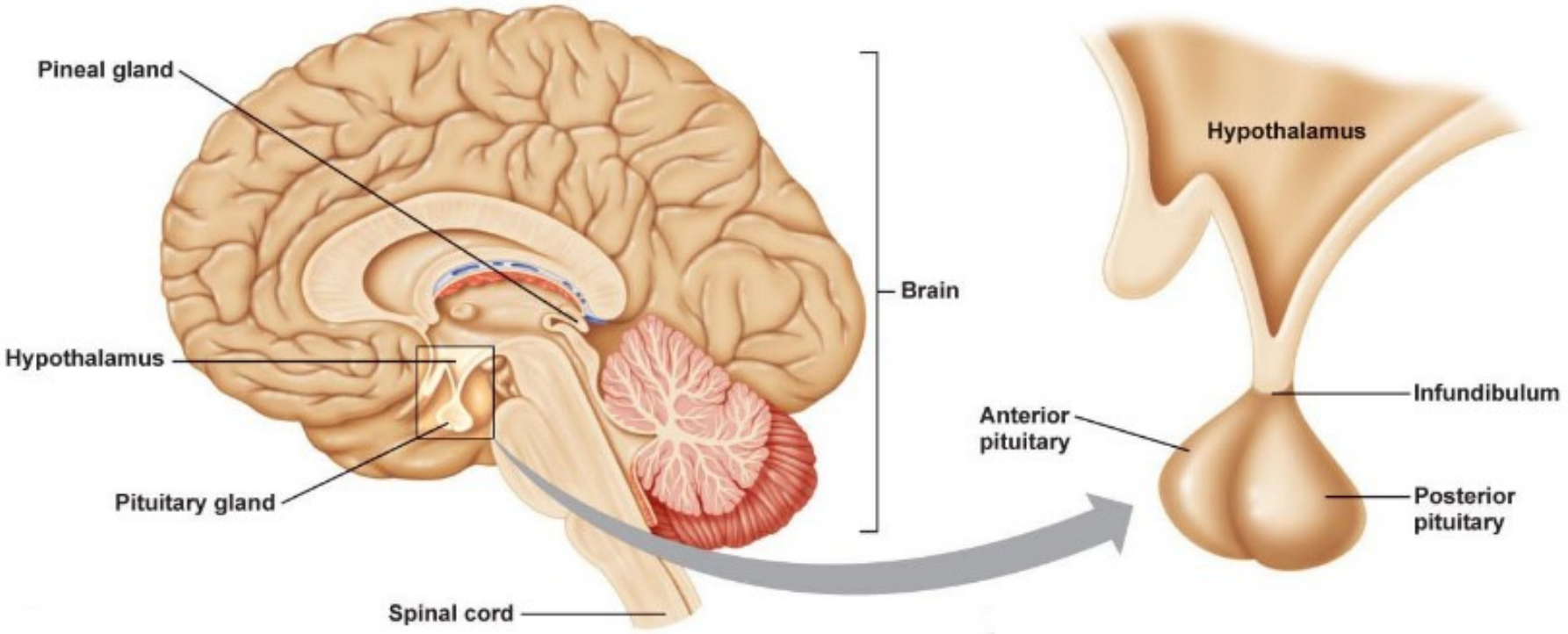
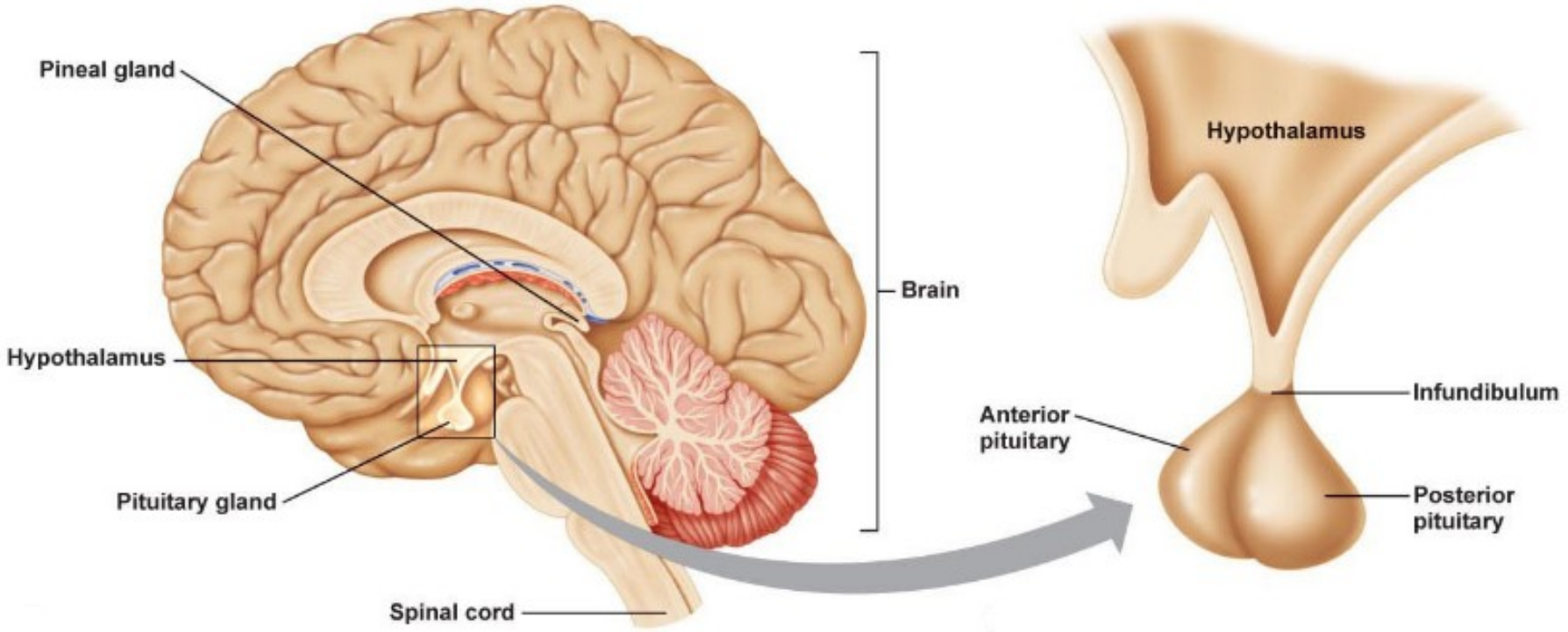
The hypothalamus and pituitary



Chemical Messengers

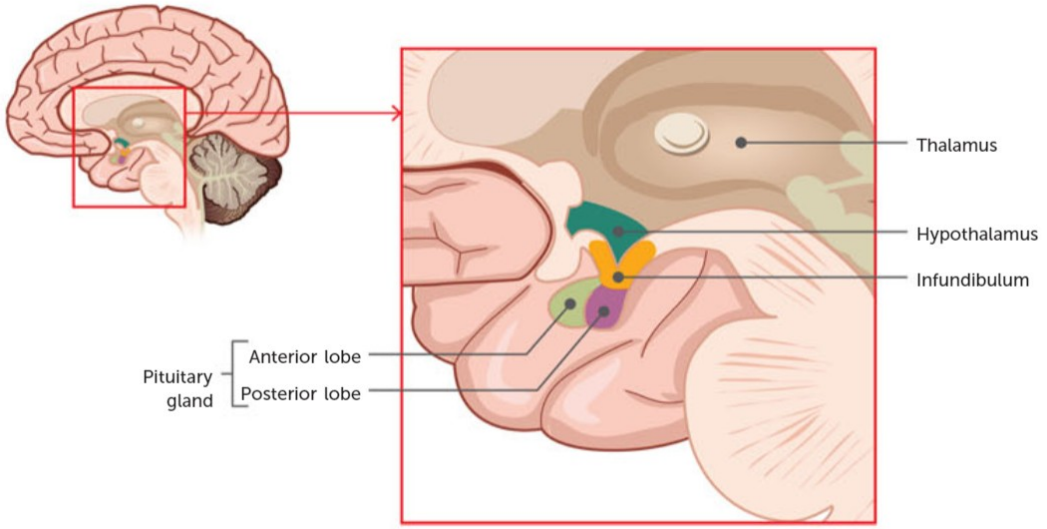
The Hypothalamus and Pituitary

- Work together to control many of the other endocrine glands – hormone homeostasis
- Located in brain:



The Hypothalamus

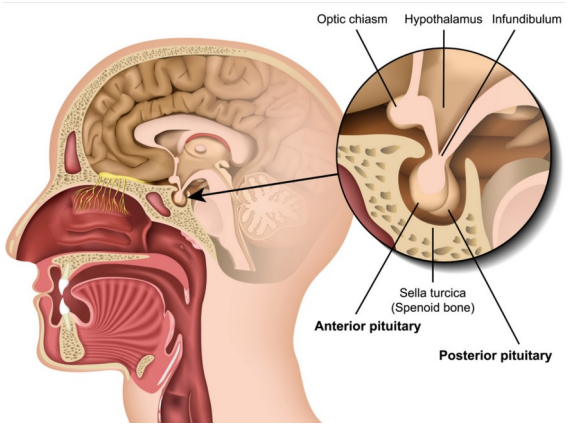
- Has roles in both the endocrine and nervous systems – acts as a connector between the two.
- Regulates many basic functions of the body:



  - ✓ Body Temperature
  - ✓ Water Balance
  - ✓ Heart Rate
  - ✓ Secretion of Releasing Factors and Inhibiting factors that stimulate the anterior pituitary to produce other hormones or to inhibit the release of hormones.
  - ✓ Production of hormones that travel along nerves and are released by the posterior pituitary.
- Located in the base of the brain, below the thalamus and above pituitary.

Learning Aim: Describe the location of the hypothalamus and its basic functions

The Pituitary Gland



- Also called the *hypophysis*, sometimes called the "master gland".
- Lies just beneath the hypothalamus in a pocket of bone.
- Joined to the hypothalamus by a stalk called the *infundibulum*.
  - ✓ The infundibulum contains blood vessels and nerves that communicate with the pituitary.
- Two lobes that function separately:

  - ✓ Anterior Lobe – closest to front of body.
    - Receives releasing factors from the hypothalamus via blood vessels in the infundibulum.
    - Produces hormones in response to stimulus by hypothalamus and releases them into *systemic circulation* (the blood stream supplying the body).

  - ✓ Posterior Lobe – closest to back of body.
    - Does not make its own hormones.
    - Releases hormones made by the hypothalamus that travel down the infundibulum via nerves.

Learning Aim: Describe the structure and location of the pituitary gland.

Communication between Hypothalamus and Anterior Pituitary

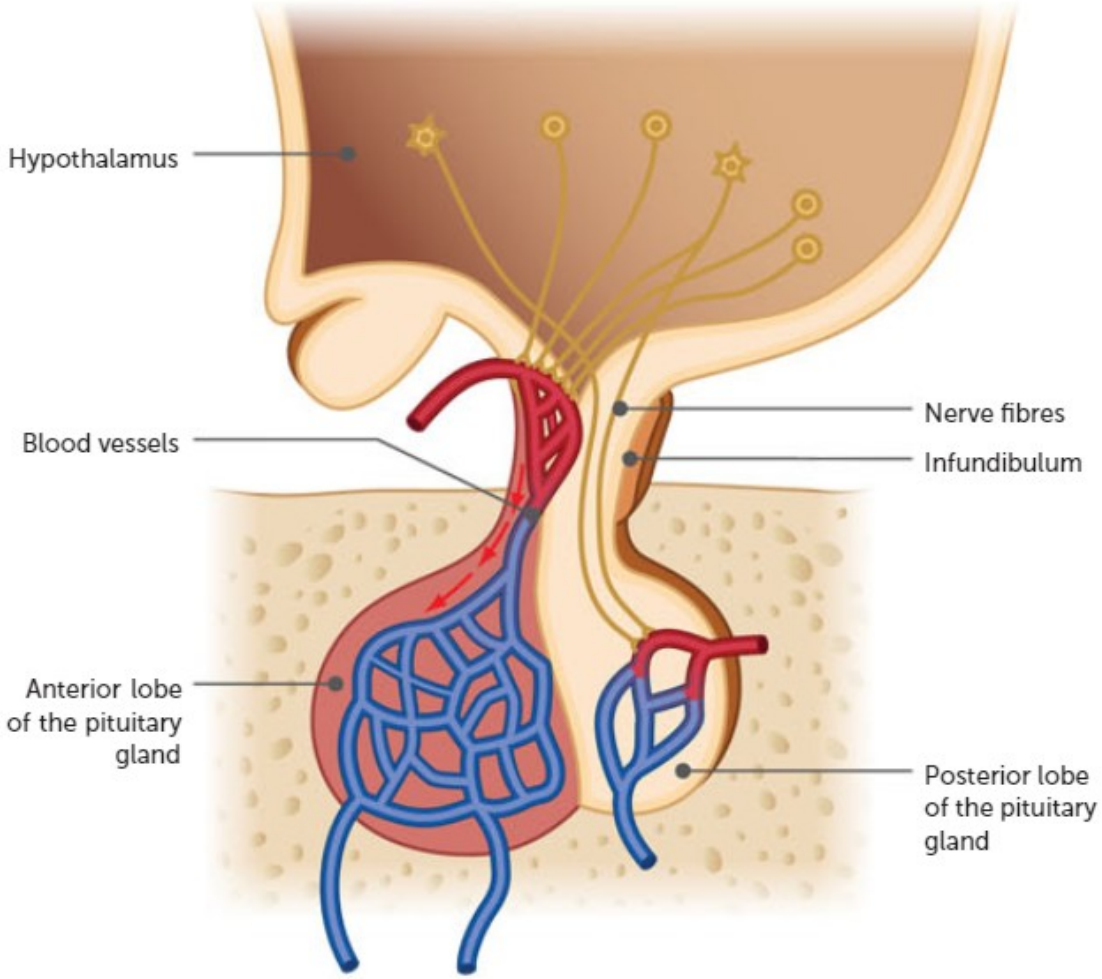
1: the hypothalamus produces releasing factors.

2: these are secreted into the blood vessels of the infundibulum.

3: releasing factors travel through these blood vessels to the anterior pituitary.

4: the releasing factors stimulate the anterior pituitary to produce and release other hormones.

5: the hormone produced is released into the systemic circulation



Releasing Factors from the Hypothalamus Include:

- TRF: Thyrotropin Releasing Factor
- CRF: Corticotropin Releasing Factor
- GHRF: Growth Hormone Releasing Factor
- PRF: Prolactin Releasing Factor
- GnRF: Gonadotropin Releasing Factor

Inhibiting Factors from the Hypothalamus Include:

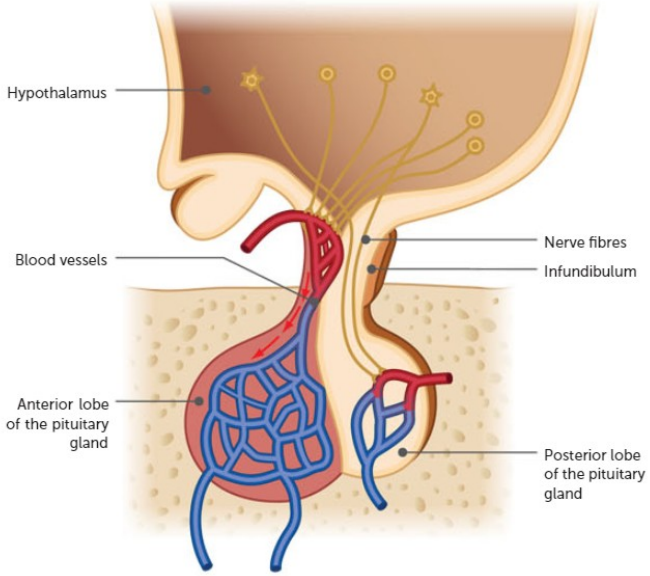
- PIF: Prolactin Inhibiting Factor (also called Dopamine)
- GHIF: Growth Hormone Inhibiting Factor (also called Somatostatin)

Learning Aim: List the releasing factors that are produced by the hypothalamus
Learning Aim: Describe how the releasing hormones travel from the hypothalamus to the anterior pituitary.

The Anterior Pituitary

- Also called the *adenohypophysis*
- Produces and releases a number of hormones in response to releasing factors from the hypothalamus.

Hypothalamus Produces:		In response, the Anterior Pituitary
TRF Thyrotropin Releasing Factor	Releasing/ Inhibiting factors travel via the blood vessels of the infundibulum to the Anterior Pituitary	Produces and releases **TSH: Thyroid Stimulating Hormone** into the systemic circulation
CRF Corticotropin Releasing Factor		Produces and releases **ACTH: Adrenocorticotropic Hormone** into the systemic circulation
GHRF Growth Hormone Releasing Factor		Produces and releases **Growth Hormone** into the systemic circulation
PRF Prolactin Releasing Factor		Produces and releases **Prolactin** into the systemic circulation
GnRF Gondaotropin Releasing Factor		Produces and releases **LH: Luteinising Hormone** and **FSH: Follicle Stimulating Hormone** into the systemic circulation
PIF Prolactin Inhibitory Factor		Inhibits release of Prolactin
GHIF Growth Hormone Inhibitory Factor		Inhibits release of Growth Hormone



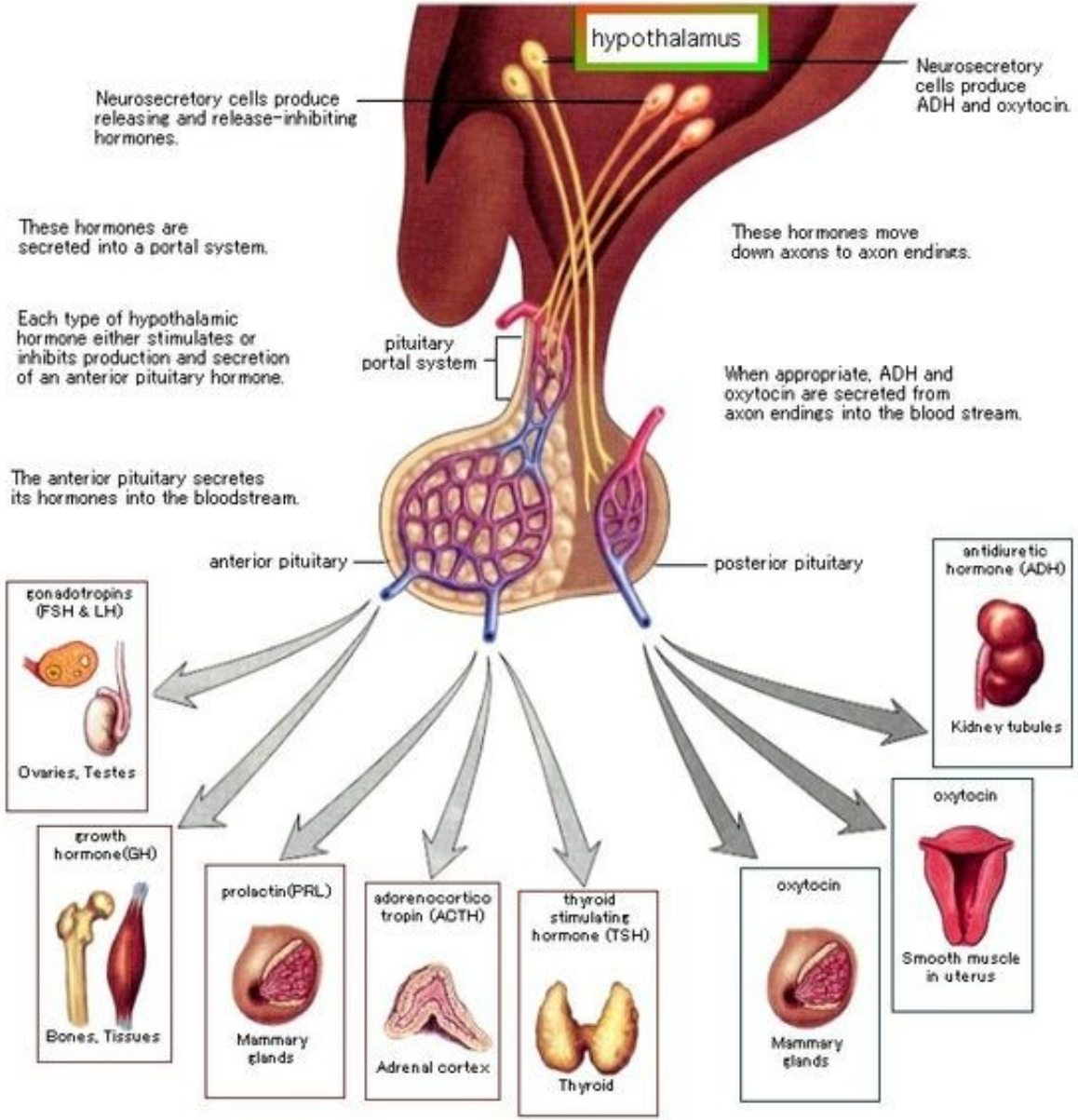
Learning Aim: Link the releasing factors from the hypothalamus to the anterior pituitary hormones they stimulate.

Anterior Pituitary Hormone Functions:

HORMONE	TARGET ORGAN	MAIN EFFECTS
Anterior lobe of the pituitary gland		
Follicle-stimulating hormone (FSH)	Ovaries (females) Testes (males)	Growth of follicles Production of sperm
Luteinising hormone (LH)	Ovaries (females) Testes (males)	Ovulation and maintenance of corpus luteum Secretion of testosterone
Growth hormone (GH)	All cells	Growth and protein synthesis
Thyroid-stimulating hormone (TSH)	Thyroid gland	Secretion of hormones from the thyroid
Adrenocorticotropic hormone (ACTH)	Adrenal cortex	Secretion of hormones from the adrenal cortex
Prolactin (PRL)	Mammary glands	Milk production

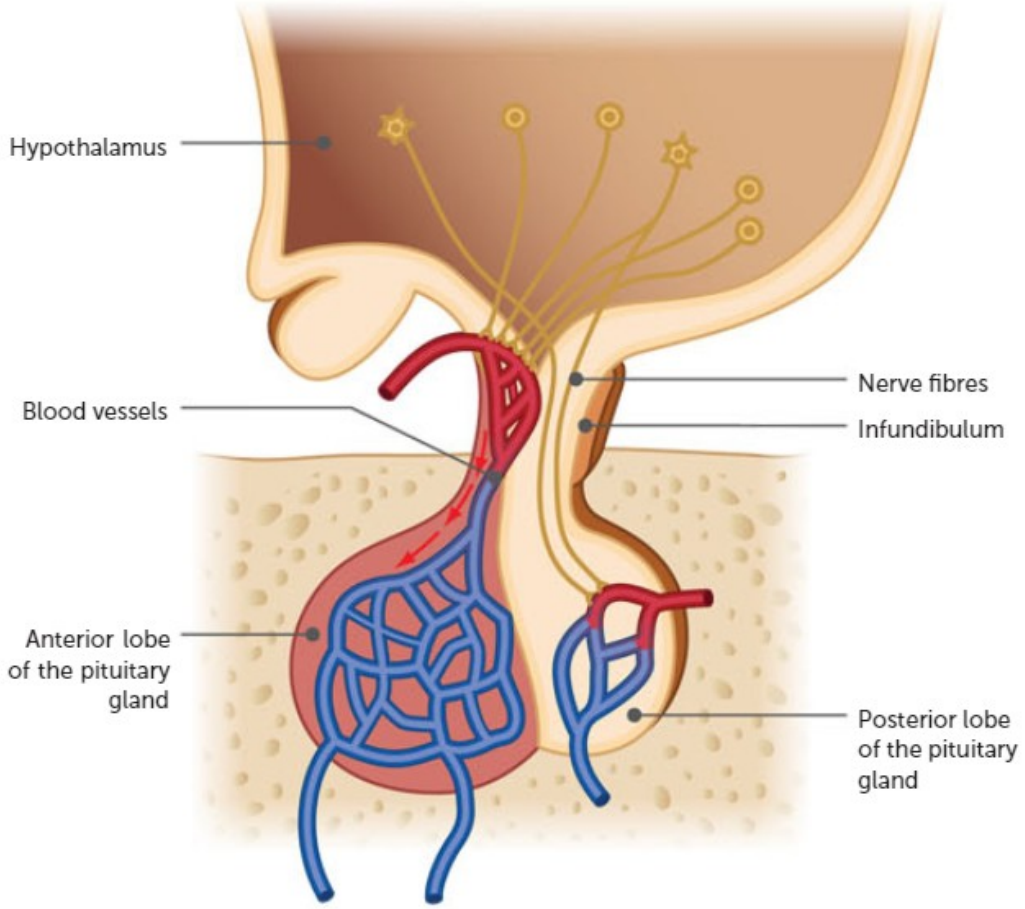
Learning Aim: Describe the functions of the hormones produced and released by the anterior pituitary

Anterior Pituitary Hormone Functions:



The Posterior Pituitary

- Also called the *neurohypophysis*.
- Does not manufacture hormones of its own.
- Hormones are made in the hypothalamus and travel through nerve fibre extensions in the infundibulum and are released from the posterior pituitary in response to nerve impulses.



Learning Aim: Describe the functions of the posterior pituitary gland.

Communication between Hypothalamus and Posterior Pituitary

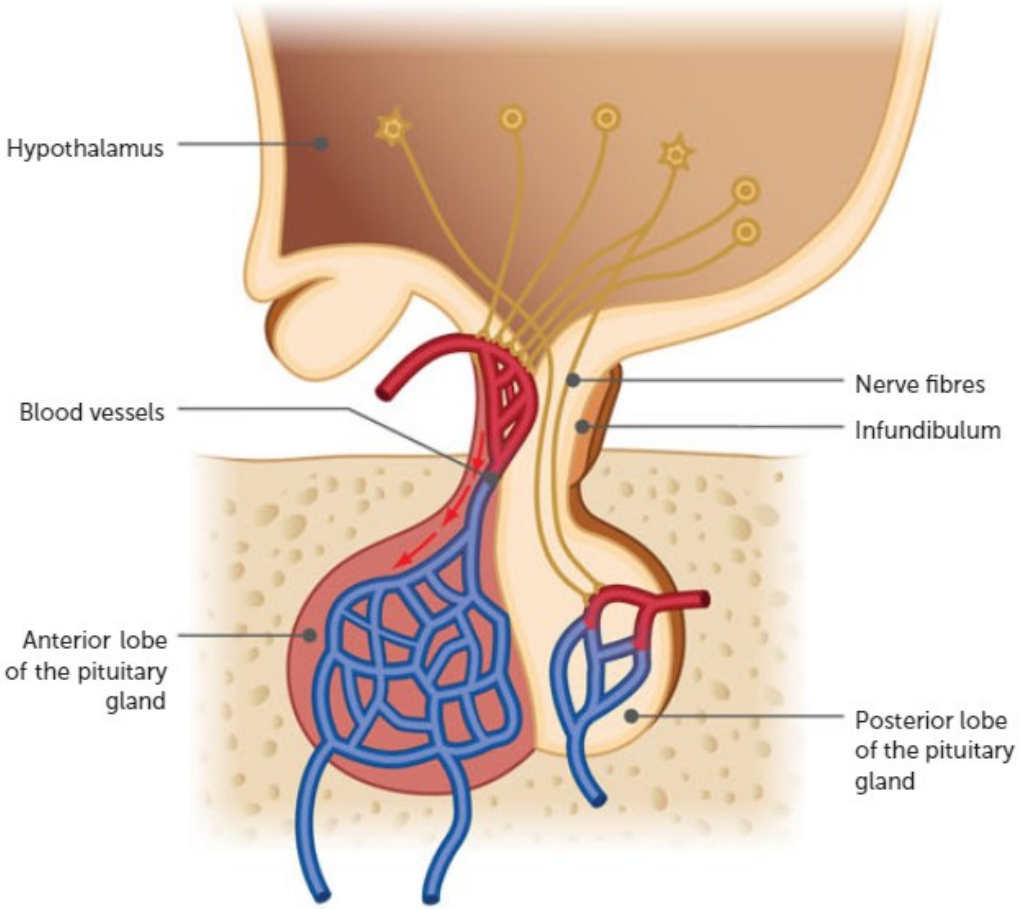
1: the hypothalamus manufactures the hormones *oxytocin* and *ADH: Anti-Diuretic Hormone*

2: these travel down the nerve fibres within the infundibulum to the posterior pituitary.

3: The hormones are then released into the systemic circulation by the posterior pituitary in response to nerve impulses.

Posterior Pituitary Hormones and Actions:

- Oxytocin
  - ✓ Stimulates uterine contractions.
  - ✓ Stimulates milk ejection from mammary glands.
- Anti-Diuretic Hormone
  - ✓ Also called vasopressin.
  - ✓ Causes kidneys to reabsorb water from urine that is forming, and return it to the blood stream – helps to retain fluid in the body.
  - ✓ Can also cause constriction of arterioles at high concentrations.



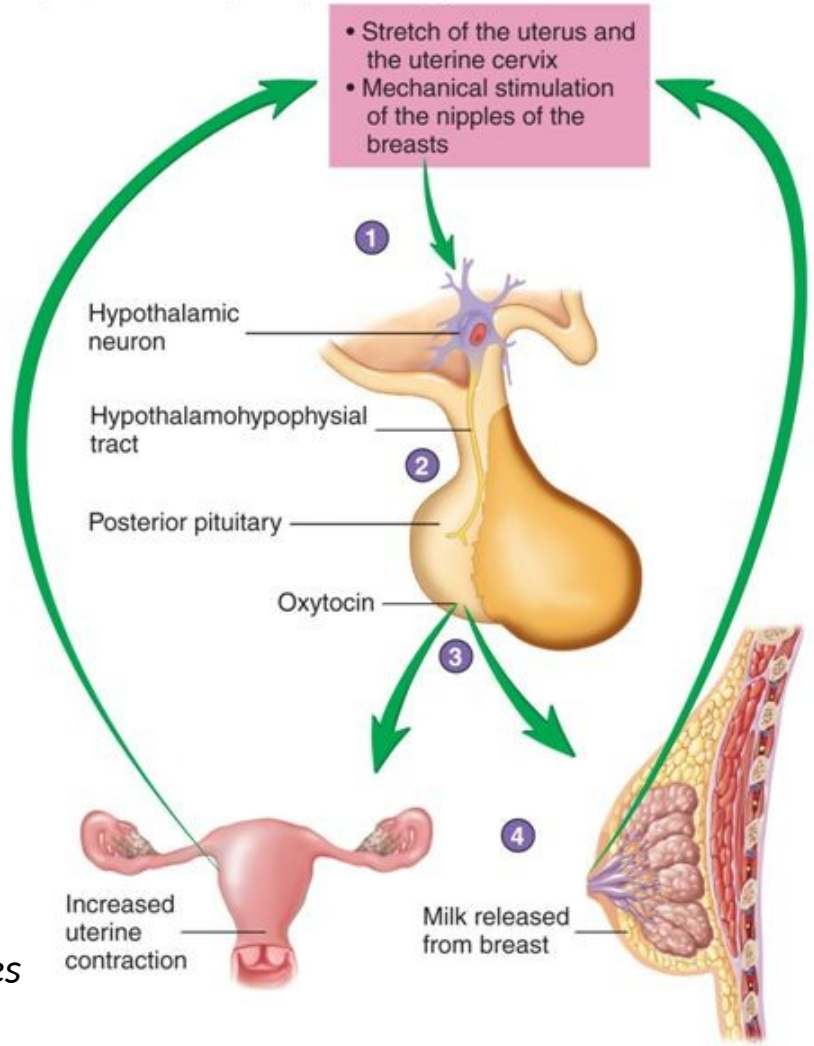
Learning Aim: Describe the functions of the posterior pituitary gland.
Learning Aim: Comment on where the hormones of the posterior pituitary are manufactured.

Regulation of Posterior Pituitary Hormones

Control of Oxytocin Secretion

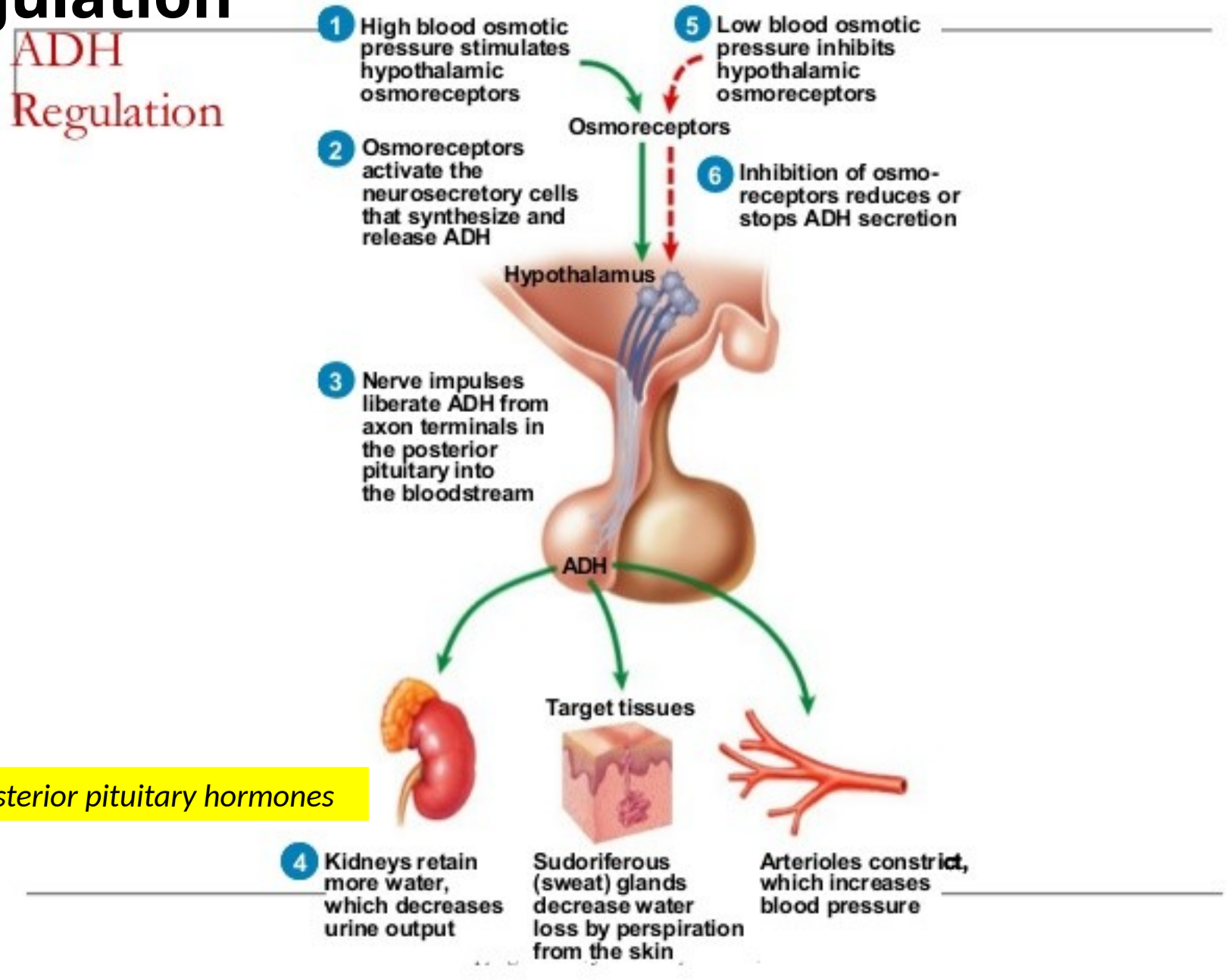
Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

1. Stretch of the uterus and the uterine cervix or stimulation of the breasts' nipples increases action potentials in axons of oxytocin-secreting neurons.
2. Action potentials are conducted by sensory neurons from the uterus and breast to the spinal cord and up ascending tracts to the hypothalamus.
3. Action potentials are conducted by axons of oxytocin-secreting neurons in the hypothalamohypophysial tract to the posterior pituitary, where they increase oxytocin secretion.
4. Oxytocin enters the circulation, increasing contractions of the uterus and milk ejection from the lactating breast.



Learning Aim: Describe the regulation of posterior pituitary hormones

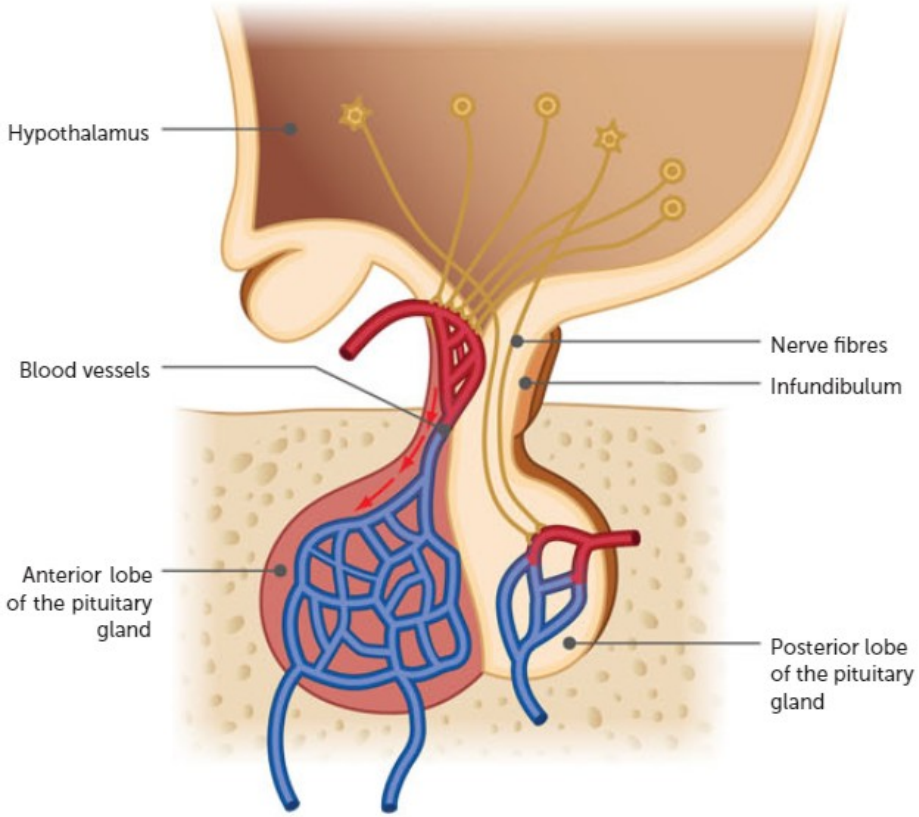
Regulation of Posterior Pituitary Hormones
Example: ADH regulation



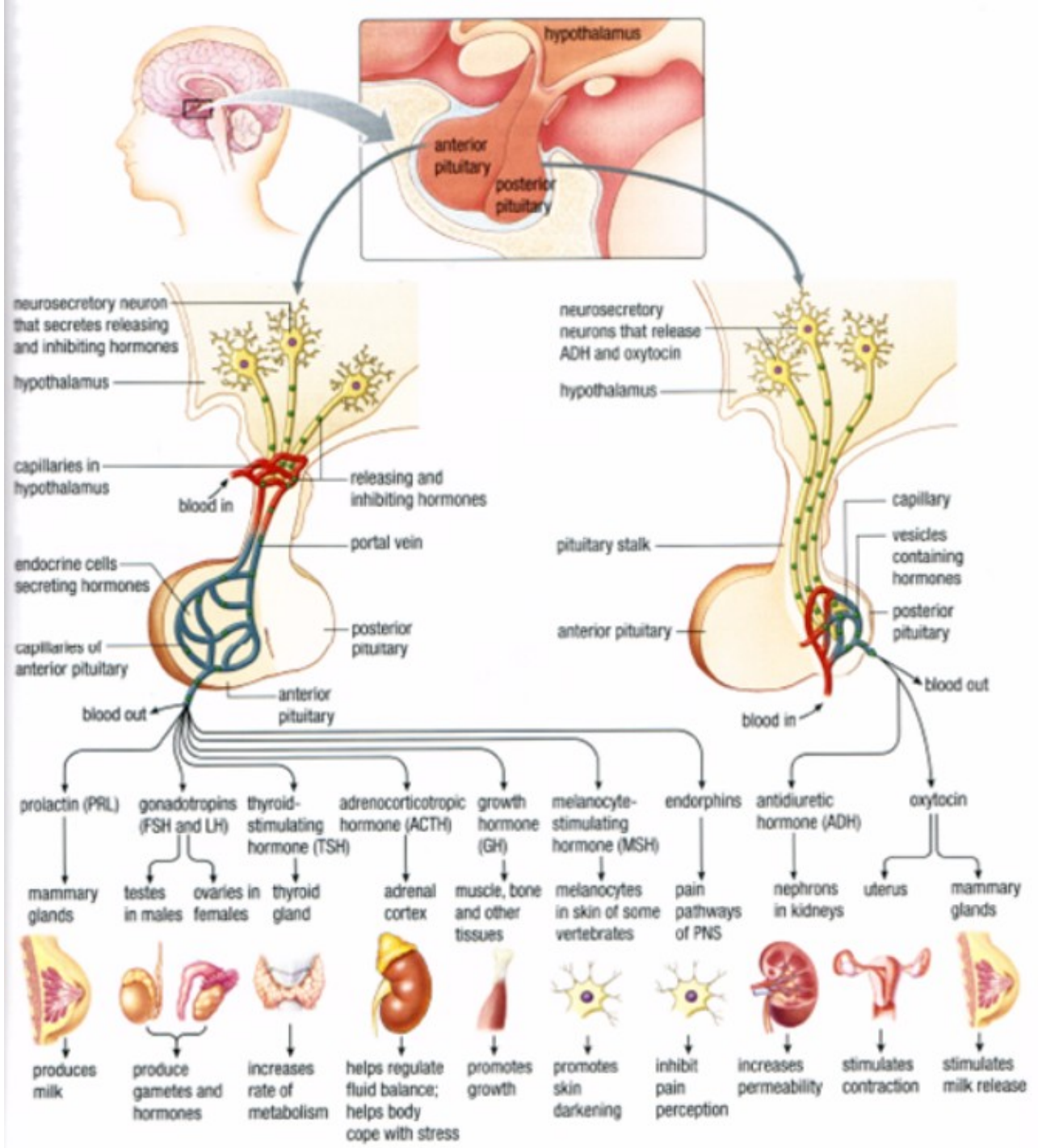
Learning Aim: Describe the regulation of posterior pituitary hormones

Compare Anterior and Posterior Pituitary

	Anterior Pituitary	Posterior Pituitary
Location	Beneath Hypothalamus	Beneath Hypothalamus
Connection to Hypothalamus	Blood Vessels in the Infundibulum	Long nerve fibres running through the infundibulum
Structure	Secretory cells	Nerve fibres from hypothalamus
Makes hormones	Yes, lots of them	No, just releases hormones made in the hypothalamus
Stimulated by	Releasing Factors from hypothalamus	Nerve signals in the fibres from the hypothalamus
Releases Hormones	Into systemic circulation	Into systemic circulation



Learning Aim: Comment on the differences between the anterior and posterior pituitary.


hypothalamus
anterior pituitary
posterior pituitary
neurosecretory neuron that secretes releasing and inhibiting hormones
hypothalamus
capillaries in hypothalamus
blood in
releasing and inhibiting hormones
portal vein
endocrine cells secreting hormones
capillaries of anterior pituitary
posterior pituitary
anterior pituitary
blood out
neurosecretory neurons that release ADH and oxytocin
hypothalamus
capillary
vesicles containing hormones
pituitary stalk
posterior pituitary
anterior pituitary
blood out
blood in
prolactin (PRL)
gonadotropins (FSH and LH)
thyroid-stimulating hormone (TSH)
adrenocorticotropic hormone (ACTH)
growth hormone (GH)
melanocyte-stimulating hormone (MSH)
endorphins
antidiuretic hormone (ADH)
oxytocin
mammary glands
testes in males
ovaries in females
thyroid gland
adrenal cortex
muscle, bone and other tissues
melanocytes in skin of some vertebrates
pain pathways of PNS
nephrons in kidneys
uterus
mammary glands
produces milk
produce gametes and hormones
increases rate of metabolism
helps regulate fluid balance; helps body cope with stress
promotes growth
promotes skin darkening
inhibit pain perception
increases permeability
stimulates contraction
stimulates milk release